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A review of wetlands remote sensing and defining new considerations

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A Review of Wetlands Remote Sensing and Defining New Considerations

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Significant progress has been made in using remote sensing as a means of acquiring information about wetlands. This research provides a brief review of selected previous works, which address the issues of wetland identification, classification, biomass measurement, and change detection. Suggested new research emphases include compiling basic spectral-reflectance characteristics for individual wetland species by means of close-range instrumentation, analyzing canopies architectures to facilitate species identification, and assessing the impact on composite spectral signatures of wet soils and variable depths of standing water beneath emergent canopies. These research foci are justifiable when considered in the context of environmental change / variability and the production of trace gases.

Keywords: Wetlands; Biomass; Airborne sensor; Species; Canopy

1. INTRODUCTION

Wetlands are a valuable natural resource of considerable scientific interest because they are associated with biological diversity, important ecosystem functions and processes, and useful, economically viable products. These systems have been called the “kidneys” of the landscape because of their

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important hydrological and chemical functions (Mitsch and Gosselink, 1993). Wetlands provide for wildlife habitat, ground-water recharge, flood control, sediment filtration, nutrient retention, pollutant removal, and wetland products, such as decomposed plant matter and peat, fish, rice, and some forest resources (Dugan, 1993). Attention by researchers has also focused on wetlands because standing waters are sites of important biogeochemical processes such as the production of methane, nitrous oxide, and carbon dioxide, as well as the fixation of carbon and nitrogen (Mathews and Fung, 1987; Harden and Chanton, 1994; Verma *et al.*, 1995). Historically, wetlands were often regarded as a hindrance to productive land use (*e.g.*, agricultural activities), and were drained, filled, or manipulated to “improve” their worth and contribution to society (Mitsch and Gosselink, 1993). Their key role in sustaining the health of an ecosystem has only recently been recognized, and over the last forty years, various state and federal agencies have placed some emphasis on mapping the spatial distributions of wetlands and classifying the systems according to their morphological characteristics and growth environments. However, mapping wetlands can be a major challenge because they are found in diverse geographic areas ranging from small tributary streams, to shrub or scrub and marsh communities (Cowardin *et al.*, 1979). Over the past three decades, remote sensing has been a powerful tool for mapping and assessing wetland environments. The objectives of this paper are to: (1) provide a review of selected approaches and findings on the subject of remote sensing of wetlands; and (2) highlight the need for new and different types of remotely sensed information and analyses as they relate to wetlands.

2. REMOTE SENSING AS A TOOL FOR ANALYZING WETLANDS

Wetland identification, mapping, and analyses are often done by means of remote sensing, a tool offering well-documented advantages including a synoptic view, multi-spectral data collection, multi-temporal coverage, and cost effectiveness. In addition, anyone who has conducted field research in wetlands is well aware of the difficulty in making *in-situ* measurements. The technology has been used extensively because remote sensing seems to be one of the only practical ways to study environments that can be difficult to access or traverse. Remote sensing of wetlands can be accomplished using a

wide range of instruments and sensor systems such as aerial photography, airborne multispectral scanners, satellite imagery, and ground-based measurements using spectrometers.

3. SELECTED EXAMPLES OF PREVIOUS RESEARCH

3.1. Aerial Photography

Several studies have used air photos as the principal tool for identifying, mapping, and/or analyzing wetlands. For example, Seher and Tueller (1973) used color and color-infrared (IR) photos to map marsh vegetation in four wildlife preserves in Nevada, and determined that both types of film were equally useful for identifying and mapping wetland vegetation. More important than type of film used, however, was the scale of the photography. Not surprisingly, photographs acquired at a scale of 1:1,000 were superior to 1:10,000 because of the detail that could be derived from them. McEwen *et al.* (1976), who used aerial photographs to investigate procedures for mapping coastal wetlands near Sapelo Island, Georgia, noted the importance of orthophotoquads for the mapping project, and also suggested that color-infrared air photos be used for establishing wetland boundaries. Klemas *et al.* (1974) used electronic enhancement of high-altitude color-IR air photos to successfully classify and map wetlands in Delaware Bay. Stewart *et al.* (1980), who reported on an experiment conducted in Florida, tested three different classification schemes, based on interpretation of both black-and-white and color-IR aerial photography, and concluded that the "Cowardin system", later used for the National Wetlands Inventory (see below), was best. Carter *et al.* (1979) tested the utility of seasonal color-infrared photographs for mapping wetlands in the Tennessee Valley Region (TVR). A classification scheme, which was developed specifically for the TVR, was used during the photointerpretation process. The authors concluded that the seasonal color-IR photography provided adequate detail to map wetland areas as small as 0.5 ha and 20-m width. Using color-IR photographs recorded on six dates between 1972 and 1985, Welch *et al.* (1988) performed a comprehensive assessment of vegetation growth and water quality for Lake Marion, SC. A geographic information system (GIS) was used to analyze the aerial photo data in conjunction with water quality information for monitoring the growth and distribution of aquatic vegetation.

3.2. Airborne Multispectral Imaging

Some researchers have used non-photographic airborne sensors to analyze wetlands. Jensen *et al.* (1984) examined data collected over the Savannah River marshes in South Carolina by means of an airborne Daedalus-1260 multispectral (11-channel) scanner (MSS). The data were resampled to a nominal spatial resolution of 2.8 meters, and supervised pattern recognition procedures implemented to derive a wetland thematic map. Five categories of wetland and upland vegetation were derived, and classification accuracy was judged to be 83% in one of the study sites. The transformed divergence algorithm was used to identify the optimum bands that would provide good spectral separability between various wetland and upland vegetation including emergent marsh, scrub-shrub, mixed deciduous swamp forest, and mixed deciduous upland forest.

In another study, Jensen *et al.* (1986) combined airborne MSS data, including a thermal-infrared channel, with aerial photography and Landsat Thematic Mapper (TM) imagery to map a variety of wetland conditions along the Savannah River watershed. The airborne MSS data were used to derive the type and spatial distribution of wetland vegetation found in individual creek deltas within the floodplain. The wetland vegetation information was correlated with water temperatures using the MSS thermal data to determine the specific types of vegetation present at varying thermal gradients. Jensen *et al.* (1994) also used airborne MSS data to perform a wetland change detection study documenting revegetation, between 1981–1992, in the Savannah River swamp. Because the swamp was the recipient of thermal effluent from the US Department of Energy's (USDOE) Savannah River Site (SRS), much of the original wetland vegetation had been altered. After 1985, thermal effluent was not allowed to enter some areas of the swamp, and resulted in significant vegetation regeneration, as was evidenced from 1992 airborne MSS data.

Phinn *et al.* (1999) investigated the feasibility and accuracy of hand-held and airborne remotely sensed data to estimate vegetation structural parameters for Pacific cordgrass (*Spartina foliosa*) in a restored wetland area in southern California. Their fundamental premise was that any application of remotely sensed data to a monitoring problem should be based on establishing a link between the information required and the measurements derived from remote sensing. Airborne multispectral images were acquired using the Airborne Data Acquisition and Registration (ADAR) system, and a comparative evaluation performed with ground-based radiometric and biophysical data. The conclusions revealed that several of the biophysical

properties of coastal wetland vegetation could be estimated from high spatial resolution airborne and ground-based multispectral data.

3.3. Satellite Multispectral Imaging

Numerous investigators have worked on developing techniques for using satellite data in wetlands classification and mapping. Their research includes the use of Landsat-MSS, TM and SPOT multispectral data for these purposes. For example, Work *et al.* (1974) focused on the wetlands of North Dakota to evaluate Landsat MSS data as an alternative for, or a complement to, aerial photography. The study utilized simple level-thresholding of the MSS-7 (near-infrared) data to delineate wetlands. Work and Gilmer's (1976) paper examined the problem of "mixed pixels", and the authors estimated the fraction of a pixel that was composed of open water. Gilmer *et al.* (1980) used high-resolution aircraft imagery to adjust their wetland census derived from Landsat-MSS data. Seevers *et al.* (1974) used Landsat-MSS data to inventory the wetlands of the Nebraska Sand Hills. Manual interpretation of photographic enlargements of early-season and late-season MSS band 5 (visible red) and band 7 (near-infrared) images allowed classification of wetlands, ten acres or more in size, into four general groups (open water, sub-irrigated meadow, marsh, seasonally flooded basins) at an accuracy of 85%. Turner and Rundquist (1981) used digital processing of over 100 Landsat-MSS images to identify, map, and inventory wetlands in all or parts of nine states in the North-Central U.S. for the Army Corps of Engineers. Overlay maps delineating wetland areas were prepared at a scale of 1 : 24,000 (USGS topographic quadrangles). Areas of open water, marshes, wet meadows, and riparian wetlands were measured and summarized in a tabular format on a per quad basis.

Few researchers have made use of thermal-infrared sensors in studying wetland systems. One exception was Queen *et al.* (1984) who used the NASA thermal-infrared multispectral scanner (TIMS) in an attempt to distinguish among open-surficial water, marsh, and subirrigated meadows in the Sandhills of Nebraska. The six channels of TIMS data were input to a discriminant-analysis algorithm. The overall accuracy of classification was 100% for open water, 81.8% for marsh, and 87.5% for subirrigated meadow. An interesting finding was that floating aquatic vegetation was discernible in the TIMS images. In another study, Rundquist *et al.* (1985) exploited TIMS data over the same area of the Nebraska Sandhills to examine an apparent flow-through regime in Crescent Lake. The latter research served to document the linkage between ground water and lake-flow systems in the region.

Jensen *et al.* (1986) used pre-dawn data from the Daedalus-1260 multi-spectral scanner to document the thermal effluent from the Savannah River Plant into the river. The thermal data proved valuable in relating the type of wetland vegetation found in particular locales with water temperature.

3.4. Radar Imaging

The earliest use of imaging radar for mapping wetlands was motivated by the capability of such systems to penetrate cloud cover, as well as non-dependence on solar illumination (Waite *et al.*, 1981). In addition, analyses of SEASAT synthetic-aperture radar (SAR) data provided evidence that low-frequency systems (L-band) can facilitate the delineation of land-water interfaces beneath a canopy of vegetation. This unique feature has been exploited in detecting both the presence and areal extent of flooding beneath stands of vegetation (Imhoff and McCandless, 1988; Hess *et al.*, 1990). Rundquist *et al.* (1997) investigated ERS-1 and Landsat-TM data, individually and in combination, for a wetland site in the Nebraska Sand Hills. They found the TM to be superior to ERS for general wetland mapping but the ERS was better for detecting individual stands of emergent, floating, and submerged macrophytes. With the increasing availability and usage of operational SAR sensors, it has become practical to map wetland ecosystems around the world regardless of cloud or vegetation cover (Pope *et al.*, 1994; Ramsey III, 1995; Kasischke and Bourgeau-Chavez, 1997).

Although the lower microwave frequencies (*i.e.*, longer wavelengths such as P- and L-band) are useful in penetrating canopies of vegetation, higher frequencies (*i.e.*, shorter wavelengths such as X-band) have been used to detect open-surficial water (Henderson, 1995). Techniques have also been developed, using multi-frequency data; not only to measure the areal extent of water bodies but also to estimate soil-moisture levels in areas surrounding lakes (Narayanan and Hegde, 1995). Additional works on characterization of saturated soils and relative wetness include Merot *et al.* (1994) and Phillips (1990). Merot *et al.* (1994) used the European Remote Sensing (ERS-1) satellite radar system to survey saturated areas. Their results indicated that the impact of saturated soils on the backscattering coefficient is generally underestimated or neglected. To obtain a better indication of the moisture conditions, this coefficient should be given careful consideration. The study also concludes that variations in topography influence soil moisture content – *i.e.*, between hillslopes and flat terrain features.

4. TYPICAL PREVIOUS WETLAND APPLICATIONS OF REMOTELY-SENSED DATA

4.1. Inventory and Mapping

A number of wetland-classification schemes have been developed and applied in the U.S. Two of the most prominent, both programs of the Fish and Wildlife Service (USFWS), were aimed not only at locations of various types of wetlands but also at standardizing definitions, interpretations, and procedures for large inventories.

In 1954, the USFWS conducted a national survey of wetlands that were deemed important to waterfowl (Shaw and Fredine, 1956). The effort, which encompassed about 40% of the lower 48 states (Pywell and Wilen, 1991), was organized around four broad wetland types; inland-fresh, inland-saline, coastal freshwater, and coastal-saline, with subdivisions arranged in order of water depth or frequency of inundation (Mitsch and Gosselink, 1993). In all, 20 wetland types were identified, 8 of which were associated with the inland-fresh category. The source of data included air photos, ground observations, and existing maps.

The USFWS National Wetlands Inventory (NWI) was developed to provide coverage of the entire country using a hierarchical classification (Cowardin *et al.*, 1979). Data sources included air photos, soils data, and topographic maps (Lyon, 1993; Dugan, 1993). The system describes ecological taxa, arranges them in a system useful for resource managers, and provides uniformity of concepts and terms. At the highest level of the hierarchy, there are five wetlands and deepwater habitats including marine, estuarine, riverine, lacustrine, and palustrine. With the exception of the palustrine systems, the other wetland habitats are further divided into subsystems. Marine and estuarine systems each have two subsystems: subtidal and intertidal; the riverine systems are comprised of tidal, lower perennial, upper perennial, and intermittent subsystems; and the lacustrine subsystems include limnetic and littoral.

4.2. Measuring Vegetative Biomass

A considerable amount of work, some more than 20 years old, has been done on the subject of measuring wetland biomass by means of remote sensing, although much of that work has been done in coastal salt-marsh environments (as opposed to inland fresh systems). For example, Reimold *et al.* (1973), who focused on primary production of tidal marsh near Sapelo

Island, Georgia, used color-infrared aerial photography at various scales (and also airborne thermal-infrared scanning) to successfully estimate biomass and other vegetative parameters. Drake (1976) used a Tektronic J-16 two-band radiometer to measure red (656–705 nm) and near-infrared (776–826 nm) spectral reflectances for estimation of standing biomass in three salt-marsh communities throughout the growing season. He concluded that the inverse of red reflectivity was a good predictor of green biomass in sedge (mixture of *Scirpus olneyi* with some *Spartina patens* and *Distichlis spicata*) and grass communities (mixture of *Spartina patens* and *Distichlis spicata*), but not in a shrub community (dominated by *Iva frutescens*). Bartlett and Klemas (1981) used a “Radiant Power Measuring Instrument” and wavebands identical to the four Landsat-MSS channels to collect spectral data in the coastal marshes of Delaware. Their work focused on *Spartina alterniflora*, *Spartina patens*, and *Distichlis spicata*. Important findings included the variability of reflectance being caused by changes in canopy morphology and composition; near-infrared canopy reflectance being somewhat sensitive to vegetation amounts but also controlled by growth form; and the near-infrared/red reflectance ratio being sensitive to changes in green biomass of *S. alterniflora*. Hardisky and colleagues (*e.g.*, Hardisky *et al.*, 1983) extended their work on *S. alterniflora* biomass using the prototype Goddard Space Flight Center (GSFC) Mark-II three-band radiometer (Tucker *et al.*, 1980) configured to simulate bands 3, 4 and 5 (red, near-infrared, mid-infrared) of the Landsat-4 TM. They concluded that spectral-radiance indices correlated highly with live leaf biomass, percent live biomass, and a live-dead biomass ratio (see also Hardisky *et al.*, 1984). Similar work was conducted by Gross *et al.* (1986), who also used the GSFC Mark-II configured to simulate TM bands 3, 4, and 5 for measuring biomass of *Spartina anglica* in salt marshes in France. Prediction of biomass from spectral data, and vegetation indices was possible, although dead material in the canopies was problematic. Gross *et al.* (1987) used Landsat-TM data to effectively measure biomass of *Spartina alterniflora* in Delaware Bay.

Ramsey and Jensen (1996) examined remote sensing as tool to describe the spectral and structural changes within and between mangrove species and community types. Their study focused on areas along the southwest Florida coast, where four species of mangroves occurred. *In situ* information on canopy closure and LAI was collected to examine canopy structure, while measurements of canopy spectra were derived from helicopter platforms. Reflectance values were generated from the canopy spectral data to correspond with Landsat TM bands 1–4, SPOT XS bands 1–3, SPOT

Panchromatic, and AVHRR bands 1 and 2. Correlation and regression analyses were performed to determine relationships between canopy structure and reflectance. Their results indicated that it may be difficult to discriminate mangrove species based on optical properties alone, however, estimates of vegetative biomass such as LAI may be derived using a simple transform of the red and near-infrared bands available on various sensor systems.

4.3. Detection of Change

Over a 200-year period (1780s to 1980s), the conterminous United States lost 53 percent of its wetlands to agricultural, residential, and/or commercial land use (Dahl, 1990), and in a nine-year period (mid-1970s to mid-1980s), inland wetland for the conterminous United States declined by 2.5 percent (Dahl and Johnson, 1991). Given this situation, it is critical that remote sensing methods be used to monitor the changes in this precious natural resource. Examples of such work include Lyon and Greene (1992), who used air photos to measure changes between 1935 and 1980 in the wetlands along one shoreline of Lake Erie. Jensen *et al.* (1993a) used multi-temporal Landsat-TM data to examine two study areas in coastal South Carolina. The authors described the image-processing techniques used to produce accurate change measures. Image algebra change detection techniques were used by Jensen *et al.* (1993b) on SPOT 10-meter panchromatic imagery to delineate and quantify the seasonal change in waterlilies. Jensen *et al.* (1995) also used Landsat Multispectral Scanner (MSS), SPOT High Resolution Visible (HRV) data, and digital-processing techniques to measure change in cattail and sawgrass/cattail-mixture communities in the Florida everglades. These image-maps were then compared to the distribution of porewater phosphorus in the area. Mouat *et al.* (1993) provides a general summary of change-detection techniques and procedures for use with remote sensing data.

5. THE NEED FOR NEW AND DIFFERENT TYPES OF INFORMATION AND ANALYSES

The inventory, classification, and monitoring of changes in the spatial distribution or condition of wetlands over time remain important activities in their own right, but scientific interest in global or environmental change provides an additional rationale for surveying freshwater resources.

Wetlands, among the most productive of all ecosystems, are thought to be the major source of atmospheric methane (Valiela, 1991), which is a very effective “greenhouse gas”, and one that is increasing on a global scale (Schindler, 1991). Wetlands are efficient producers of methane (CH_4) because of their characteristic anaerobic conditions, high levels of vegetative productivity, and the bacterial action, which decomposes those organic materials. Some of the variables which are important in the emission of CH_4 into the atmosphere include flooding and soil-water content, temperature, type and amount of organic materials, vegetation, and redox potential (Prieme, 1994).

Because of limitations in the spectral and spatial characteristics of most currently available remote sensors, it has been possible to infer only the most general characteristics of wetlands from spectral information. In light of particular needs with regard to wetland inventory and change detection, including assessments of environmental variability, it would appear that there are presently some fairly specific voids in the scientific literature on the subject of remote sensing of wetlands. Specifically, these issues include (a) detailed spectral characterization of individual species, (b) assessing the impact of water beneath the vegetation canopy (on the spectral signature as well as the spatial distribution of standing water and wet soil), and (c) characterizing the architecture of the vegetative canopy. Figure 1 summarizes the status of remote sensing of wetlands based on the sensor systems used and their respective applications, and includes new considerations that are designed to address fundamental issues which are critical for future remote sensing of wetlands applications.

5.1. Detailed Spectral Characterization of Individual Species

As was illustrated in the preceding paragraphs, researchers have applied remote sensing to many varied facets of wetlands identification, classification, mapping, and analysis, with most of the published papers being oriented to survey endeavors. However, there are some fundamental issues that need further research by the scientific community. There is little documentation about the basic spectral-reflectance properties of individual species of emergent macrophytes. It would be helpful to have a firm grasp on spectral signatures for the most common and/or important species at various times during their respective phenological stages, and how those signatures might differ from one species to the next. While a “taxonomic undertaking”, in large part, such information would seem prerequisite to any serious attempt at identifying or classifying wetlands using remotely sensed data.

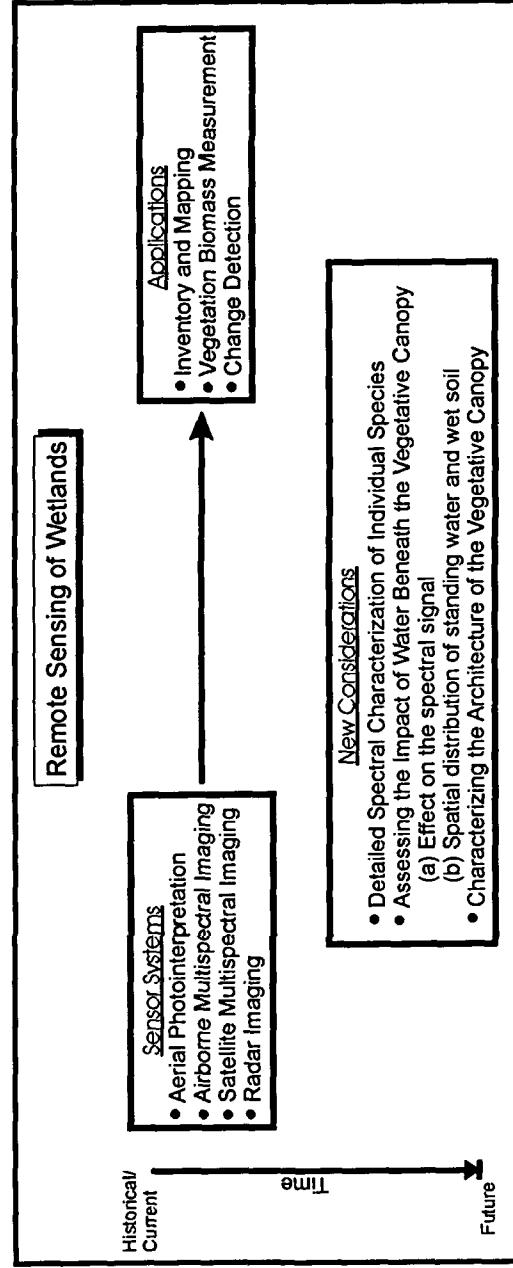


FIGURE 1 Based on the literature review, this schematic diagram illustrates the historical and current status of wetlands remote sensing. The diagram includes new considerations that are designed to address fundamental issues, which would be critical for future remote sensing of wetlands.

The earliest effort on the reflectance properties of individual wetland species was that of Anderson (1970) who worked on selected wetlands in Chesapeake Bay. He used an ISCO Model SR spectroradiometer, operating between 0.4 and 1.35 micrometers. While there was no mention of the number of spectral channels provided by the system, the discontinuous nature of the spectral curves in the publication suggests that there were 18 to 20 individual bands. Spectral-reflectance measurements of ten marsh-plant species including yellow waterlily (*Nuphar advena*), arrow arum, pickerelweed, sweet flag, wildrice (*Zizania aquatica*), cattail, reed, rush (*Juncus sp.*), and salt marsh grass were recorded. However, spectral curves for only annual wildrice (*Zizania aquatica*), waterlily (*Nuphar advena*), and rushes (*Juncus sp.*) were presented. The spectral curves revealed only minor differences from one species to the next in the visible spectrum, but greater differences in the near-infrared. The findings on spectral reflectance of marsh plants were summarized as follows:

Comparison of spectral reflectance curves indicates that narrow bandwidths around 0.9 and 1.1 micrometers would be best for separation or delineation of all species studied. At 0.9 micrometers, only reed and wild rice cannot be separated. If it is desirable to separate sweet flag, pickerelweed, wild rice and reed, all of which have medium reflectance, 0.75, 0.85 and 1.1 micrometers are good wavelengths at which to work. Low reflectors such as cattail, rush and salt marsh grass separate well at a number of wavelengths – 0.55, 0.675, 0.75, 0.8, 0.85, 0.9 and 1.3 micrometers. Strong reflectors, yellow water lily and arrow arum, should be easily delimited at 0.75, 0.8, 0.9 and 1.1 micrometers.

The author also mentions that there was variation in reflectance with changing seasons, but only a single graphic was presented that depicted spectral curves of reeds (*Phragmites*) for July and October.

Similarly, Reimold *et al.* (1973) did laboratory measurements of reflectance of rushes (*Juncus roemerianus*) and both tall and short cordgrass (*Spartina alterniflora*) with a Beckman DK-2A spectrophotometer operating in the spectral range 0.5 to 2.5 micrometers. They presented a graphic showing spectral curves for cordgrass (both forms) and the rushes, and the striking aspect of the graphic was the fact that the rushes had the lowest visible reflectivity but clearly the highest near-infrared reflectance (approximately 70%). The emphasis of the paper, however, was on estimating primary production in marshes.

Ernst-Dottavio *et al.* (1981) took field spectral measurements using an Exotech-100 radiometer mounted on a helicopter. The targets were six

major (general) wetland types found in northeastern Indiana: shallow marsh, deep marsh, open water, shrub swamp, wooded swamp, and tamarack bog. They concluded that open water and deep marsh were relatively easily separated based on spectral reflectance, but the other four categories were much more difficult to classify. Because the Exotech acquired data in only four broad bands, it was not possible to develop useful spectral curves.

Best *et al.* (1981) also used a four-band Exotech radiometer to examine pure stands of ten hydrophyte species common to the "Prairie Pothole" region of the Dakotas, including narrow-leaved cattail (*Typha angustifolia*), soft-stem bulrush (*Scirpus validus*), river bulrush (*Scirpus fluviatilis*), common reed (*Phragmites communis*), water plantain (*Alisma plantago-aquatica*), whitetop (*Scolochloa festucacea*), bur reed (*Sparganium eurycarpum*), foxtail barley (*Hordeum jubatum*), water smartweed (*Polygonum coccineum*), and prairie cordgrass (*Spartina pectinata*). Spectral-reflectance measurements were taken during the early-emergent, flowering and early-seed, and senescent phenological stages. The number of spectral separations for species was maximized during the flower and early-seed stage where eight species were distinct, and only bur reed was not separable by reflectance during at least one phenological stage. Again, the limitations of the radiometer precluded development of useful diagnostic spectral curves.

A paper by Penuelas *et al.* (1993), who used a Spectron Engineering SE-590 spectroradiometer to analyze aquatic vegetation of Searsville Lake, California, may well be the first attempt at classifying aquatic species based on a field instrument possessing true high spectral resolution. The authors provided spectral curves for "...characteristic aquatic plants of Searsville Lake...". The species investigated included: submergent types such as Hornwort (*Ceratophyllum demersum*), water milfoil (*Myriophyllum aquaticum* and *M. verticillatum*), Muskgrass (*Chara* sp.), and pondweed (*Potamogeton illinoensis*, *P. pusillus*, and *P. foliosus*); floating types such as duckweed (*Lemna minor*) and mosquito fern (*Azolla filiculoides*); and emergent types such as marsh seedbox (*Ludwigia palustris*), smartweed (*Polygonum lapathifolium*, *P. coccineum*, and *P. punctatum*), and common cattail (*Typha latifolia*). The three general types, submerged, floating, and emergent, were easily separable using only spectral data. Commonly used vegetation indices (*e.g.*, NDVI) were useful for measuring selected parameters (*e.g.*, biomass) of the emergent species, but more importantly, the authors introduced new narrow-band indices based on derivative spectra, which may prove important in evaluating plant-physiological status.

Anderson and Perry (1996) presented diagnostic curves, acquired by an ASD field spectroradiometer operating between 400 and 900 nm, for red

maples (*Acer rubrum*) in both wet and dry sites. Their purpose was to characterize leaf reflectance at different stages of flooding, and compare results to redox potentials. Although the study was not aimed specifically at characterizing wetland species, it is relevant to remote sensing of wetlands.

5.2. Assessing the Impact of Water Beneath the Vegetative Canopy

The fact that standing water (or wet soil) forms the background for a spectral signal from wetland canopies, presents two specific problems:

5.2.1. Effect on the Spectral Signal

First, additional research is needed on the effect of the water itself on the composite optical signal from emergent-vegetation canopies. While this may appear to be a very simple issue, little (if any) documentation exists on moisture interaction with the spectral signature. In our collective haste to use remote sensing as a practical tool to map and analyze wetlands, performing basic background research may have been neglected. The water beneath the vegetative canopy can be very dynamic, changing in depth, level of turbidity and/or algal production, and spatial extent from time to time; a potential problem for those analyzing data from orbiting and aircraft-mounted sensors.

5.2.2. Spatial Distribution of Standing Water and Wet Soil

Second, measures of not only the areal extent of wet soils but also the depth of standing water are needed because both are linked to hydrostatic pressure, which governs, at least to some extent, the flux of methane to the atmosphere. Given the limitations with regard to the availability of radar data from space in the recent past, it has been virtually impossible to detect the presence of standing water or wet soils under a full canopy of emergent macrophytes.

However, with the current availability of new orbital-microwave systems such as Radarsat, ERS, and JERS, it now seems possible to address this problem.

A few investigators have used satellite synthetic-aperture radar (SAR) to assess wetland environments, although most of the work was done in forest and/or salt-marsh settings (*e.g.*, Krohm *et al.*, 1983; Ormsby *et al.*, 1985), with virtually no prior work in inland freshwater systems. Hess *et al.* (1990)

provide a useful summary of the issue of detecting flooding beneath a forest canopy, which is somewhat analogous to the problem of detecting standing water beneath a canopy of wetland vegetation. Tanis *et al.* (1994) used ERS-1 SAR data to examine biophysical conditions in a coastal tidal marsh system in the Gulf of Mexico. The authors developed a model to predict backscatter based on canopy descriptors such as leaf density, height, biomass, and also noticed differences in backscatter from vegetation between periods of low *versus* high tides. Merot *et al.* (1994) used ERS-1 SAR data and “topographic analysis” to survey saturated areas. Kasischke and Bourgeau-Chavez (1997) tested ERS-1 SAR data for analyzing wetlands in southwest Florida, and found the radar to be well suited to examining soil moisture, the presence/absence of standing water, and aboveground biomass. Thus, there seems to be potential for this type of work.

5.3. Characterizing the Architecture of the Vegetative Canopy

An additional problem facing remote-sensing specialists is that some of the information required for estimating fluxes of trace gases in wetlands is more detailed and difficult to obtain *via* remote sensing than that to which we are accustomed (*i.e.*, simply identifying, classifying, and mapping the spatial distributions of those communities). For example, it seems logical that different types of plants should contribute different amounts of litter to the organic-decomposition process. Thus, accurate identification and mapping of individual wetland-vegetation species rather than the communities as mapped in typical wetland surveys would enhance studies of trace gases (*e.g.*, methane) fluxes. Being able to accurately estimate biomass in wetland communities by means of remote sensing remains an important issue. However, it would seem useful to be able to characterize the “architecture” or roughness of the vegetation canopy, for two reasons: (i) digital classifications of plant species may be enhanced if canopy-structure information were input to clustering algorithms; and (ii) the canopy roughness is linked to the aerodynamics and measurement of fluxes above the canopy (Monteith and Unsworth, 1990). The need to quantify canopy architecture to facilitate identification of wetland species suggests that radar may be the sensor of choice.

Although radar has not been employed specifically to characterize the architecture of wetland-vegetation canopies, some results from its application to crops and forests indicate potential in this regard. Microwave reflectance from vegetation canopies is related not only to its structure but also to what lies beneath. This is especially true at lower frequencies. Since

the canopy is relatively opaque to higher frequencies, radar systems operating at these wavelengths are useful in estimating plant biomass and height (Kim *et al.*, 1980). It is also possible to estimate the leaf area index (LAI) of a crop canopy using radar, especially for LAI values greater than 0.5 (Ulaby *et al.*, 1984). At lower values of LAI, the effects of soil are apparent in the signal. Similar correlations between radar reflectance and LAI have been observed for black spruce and aspen trees (Pitts *et al.*, 1988). In using radar data for LAI measurements, Prévot *et al.* (1993) concluded that a semiempirical water-cloud model of canopy backscattering could easily be inverted to derive canopy and soil characteristics. The inverted model allowed for simultaneous estimation of surface soil moisture and LAI measurements using two radar configurations – *i.e.*, X-band for LAI and C-band for soil moisture. Durden *et al.* (1995) also used radar to examine its response to LAI in a natural herbaceous wetland. Data were acquired for several flooded rice fields in central California and results indicated that C-band VV polarized radar backscatter decreased with increasing LAI, thus establishing a relationship between vegetation architecture and radar return. The results derived from the studies discussed above, lead us to believe that radar holds potential for measuring canopy biophysical properties in wetlands.

6. SUMMARY AND CONCLUSIONS

Significant progress has been made over the years in using remote sensing as a means of acquiring information about wetlands. The emphasis of most prior work has been on the identification of wetlands using remotely sensed data, classifying them into some prescribed system, and measuring biomass. Research in coastal settings is more prevalent than in freshwater environments. Some researchers have focused on changes in the spatial distribution and/or condition over time. However, research on certain basic issues such as spectral-reflectance characteristics of wetland species, and the impacts of common environmental factors such as wet soils and standing water beneath the canopy or lodging by wind, has been limited. It seems appropriate, given the attention being aimed at global/environmental variability and related production of trace gases, to address some of these issues. One useful point of departure might well be basic research in controlled, experimental settings. While controlled experiments may not be fully representative of the “natural environment”, they allow scientists to (a) develop an appropriate environment (*e.g.*, a plot comprised solely of cattail species), (b) vary the

environmental parameters (e.g., water depth, light, density of species), and (c) repeat their experiments. Results from such experiments will provide baseline data for further investigations into the fundamental characteristics of wetlands. For example, spectral reflectance curves derived for various species can be used in classifying wetland vegetation from airborne or satellite imagery at a much greater detail than what has been currently achieved. Similarly, laboratory studies of trace gas fluxes from wetland plots may be extrapolated for larger areas to analyze the overall impact of these gases on the environment.

Wetland environments present a challenge for either *in situ* based or remote sensing studies. The identification of fundamental or basic research issues is a crucial step toward a more complete scientific understanding of these complex environments.

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